# Continuous Compounding & Euler's Constant e

Exponential & Logarithmic Functions • Standard: CCSS.MATH.HSF.LE.A.4

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**1. MATHEMATICAL CONCEPT & STACK APPROACH: Model continuous exponential accumulation and compound interest using the natural exponential function [e^x].**

Exponential growth P·e^(rt) compounds continuously by scaling a base quantity according to an inner rate-time product (r × t). Evaluating from the inside out on the stack—computing r × t first, applying [e^x], and multiplying by P—follows the exact mathematical hierarchy without parentheses.

## 2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

**Published Block Kit: Prime Factor Studio (PRIME-01 Logarithmic Blocks) & Growth Curve**

Hands-On Setup: Stack Prime Factor [2] blocks (15 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$).

**Student Prediction (Before Calculator): Invest $1,000 at 7% continuous interest for 10 years: A = 1000 × e^(0.07 × 10) = 1000 × e^0.7. Will it double? (e^0.7 ~ 2.01).**

## 3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

**Keystroke Flow:** `0` `•` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

| Key Pressed | Screen (X) | Internal 4-Level Stack Effect (T, Z, Y, X) |
|---|---|---|
| `0` `•` `0` `7` | `0.07` | Rate r = 0.07 entered in X |
| `ENTER` | `0.07` | Rate lifted to Y |
| `1` `0` | `10` | Time t = 10 entered in X |
| `×` | `0.7` | Exponent r × t = 0.70 |
| `e^x` | `2.0138` | e^0.70 = 2.01375 (Growth factor: more than doubled!) |
| `1` `0` `0` `0` | `1000` | Principal P = 1000 (growth factor lifts to Y) |
| `×` | `2013.75` | Total accumulated amount A = $2,013.75! |

**4. STORY CONTEXT (The Scottish Baron Who Doubled the Lifespan of Astronomers (John Napier), Euler's e & The Rule of 72)**

In 1614, John Napier at Merchiston Castle realized that logarithms turn tedious multiplication into simple height addition—the exact engineering principle of our Prime Factor blocks (where h = 15 × log2(p) mm)! Paired with Jakob Bernoulli's continuous compounding limit e and Fibonacci's Liber Abaci (1202), students model exponential doubling and half-life decay.

**Core Investigation (Student Warm-Up A):**

A bacteria culture starts at 500 and triples every hour. Use [y^x] to calculate population after 4 hours.

# Solving for Time: Doubling Time & Natural Log [LN]

Logarithms & Inverting Exponentials • Standard: CCSS.MATH.HSF.BF.B.5

## 1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

**Concept & Stack Approach: A logarithm is the inverse of exponentiation: it converts multiplicative scaling into additive height (ln(a·b) = ln(a) + ln(b)). To solve for time t in P·e^(rt) = A on the stack, divide A by P first, take [LN] to unwrap the exponent, and divide by r.**

Blocks & Prediction: Observe how Prime Factor block heights scale with ln(p) on the comparator lid while configuring target multiplier in Y and growth rate in X. Prediction: At 5% growth (r = 0.05), doubling time is t = ln(2) / 0.05 ~ 0.69315 / 0.05 ~ 13.86 years.

## 2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[2] [LN] [0] [•] [0] [5] [÷]

| Key Pressed | Screen (X) | Internal 4-Level Stack Effect |
|---|---|---|
| [2] | 2 | Doubling target factor 2 |
| [LN] | 0.6931 | Natural log ln(2) = 0.693147... in X |
| [0] [•] [0] [5] | 0.05 | Rate r = 0.05 (ln(2) lifts to Y) |
| [÷] | 13.86 | t = ln(2) / r = 13.86 years! |

## 3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

**Misconception 1: Multiplying the base by the exponent (computing 2^5 as 2 × 5 = 10) or reversing Base and Exponent on [y^x].**

Hands-On Block Remediation: Stack five [2] Prime Factor logarithmic blocks in the comparator lid to show that adding 5 block heights multiplies 2×2×2×2×2 = 32, and verify that Base y goes in Register Y and Exponent x goes in X before [y^x].

**Misconception 2: Applying the power [y^x] after multiplying by principal P in P(1 + r)ᵗ (computing (P·(1+r))ᵗ instead of P·((1+r)ᵗ)).**

Hands-On Block Remediation: Use the Inside-Out RPN rule: compute (1 + r)ᵗ first ([1.05] [ENTER] [2] [y^x]), then multiply by P ([1000] [×]).

### 4. EXTENSION MATHEMATICAL INSIGHT

A logarithm is the inverse of exponentiation: it converts multiplicative scaling into additive height (ln(a·b) = ln(a) + ln(b)). To solve for time t in P·e^(rt) = A on the stack, divide A by P first, take [LN] to unwrap the exponent, and divide by r.

# Exponential Growth, Compound Interest & Logarithms

Name: _____ Date: _____ Class: _____

### STUDENT REFERENCE GUIDE: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLING TIME

- Exponential growth formula: A = P × e^(r·t) for continuous growth, or A = P × (1 + r/n)^(n·t).
- Natural exponential [e^x] raises Euler's constant e to power X.
- Natural logarithm [LN] inverts exponential growth: ln(e^x) = x.

**Key Sequence:** `0` `•` `0` `7` `ENTER` `1` `0` `×` `eˣ` `1` `0` `0` `0` `×`

### Investigation 1: Calculate e^2 on the calculator to 4 decimal places.

**1. Geometric / Area / Coordinate Sketch:**

**2. Algebraic Setup & Stack Plan:**

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

**3. My Prediction:**

_____

**Final Verified Answer:**

_____

**[ ] Checked on SC-32**

### Investigation 2: Calculate doubling time for an investment growing at 8% continuous interest.

**1. Geometric / Area / Coordinate Sketch:**

**2. Algebraic Setup & Stack Plan:**

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

**3. My Prediction:**

_____

**Final Verified Answer:**

_____

**[ ] Checked on SC-32**

### Investigation 3: Compute 10^3.5 using [f:] [10^x].

**1. Geometric / Area / Coordinate Sketch:**

**2. Algebraic Setup & Stack Plan:**

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

**3. My Prediction:**

_____

**Final Verified Answer:**

_____

**[ ] Checked on SC-32**

### Investigation 4: Evaluate log10(500) using [f:] [LOG].

**1. Geometric / Area / Coordinate Sketch:**

**2. Algebraic Setup & Stack Plan:**

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

**3. My Prediction:**

_____

**Final Verified Answer:**

_____

**[ ] Checked on SC-32**

# Unit 10: Exponential Growth, Compound Interest & Logarithms — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 10–12

## 1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

**Core Investigation (Warm-Up A): A bacteria culture starts at 500 and triples every hour. Use [y^x] to calculate population after 4 hours.**

Core Solution: 40,500 bacteria. Keystrokes: [3] [ENTER] [4] [y^x] [5][0][0] [×]

**Core Check Keys:** [3] [ENTER] [4] [$y^x$] [5] [0] [0] [×]

**Extension Challenge (Warm-Up B): A radioactive isotope has decay constant k = 0.035. Find its half-life: t = ln(2) / k.**

Extension Solution: 19.80 years. Keystrokes: [2] [LN] [0][.][0][3][5] [÷]

**Ext. Check Keys:** [2] [LN] [0] [•] [0] [3] [5] [÷]

## 2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

**Investigation 1: Calculate e^2 on the calculator to 4 decimal places.**

**Expected Answer: 7.3891**

**SC-32 Verification Keys:** [2] [$e^x$]

**Investigation 2: Calculate doubling time for an investment growing at 8% continuous interest.**

**Expected Answer: 8.66 years**

**SC-32 Verification Keys:** [2] [LN] [0] [•] [0] [8] [÷]

**Investigation 3: Compute 10^3.5 using [f:] [10^x].**

**Expected Answer: 3162.28**

**SC-32 Verification Keys:** [3] [•] [5] [↶] [$10^x$]

**Investigation 4: Evaluate log10(500) using [f:] [LOG].**

**Expected Answer: 2.6990**

**SC-32 Verification Keys:** [5] [0] [0] [↶] [LOG]

### Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.